

Retail Sales Analytics & Dashboard Report

Project Overview

The objective of this project was to transform a raw retail sales dataset containing four years of sales records (2021–2024) into meaningful business insights and an interactive dashboard for decision-making.

The project was completed using Python, Pandas, Plotly, and Streamlit.

1. Data Cleaning and Preparation

The raw dataset contained inconsistencies and required preprocessing before analysis.

The following cleaning steps were performed:

- Removed duplicate records.
- Standardized date formats into a consistent datetime format.
- Converted sales and profit columns into numeric values.
- Cleaned category, segment, and region values to ensure consistency.
- Handled missing values where necessary.
- Validated quantity and sales records.
- Created a cleaned dataset for analysis and dashboard development.

These steps improved data quality and ensured accurate reporting.

2. Exploratory Data Analysis

Several analyses were performed to understand business performance:

- Monthly sales trend analysis to identify seasonality.
- Sales and profit comparison by region.
- Sales analysis by product category.
- Identification of top-performing customers.
- Calculation of key business metrics and performance indicators.

3. KPI Development

The following KPIs were calculated:

- Total Revenue
- Total Profit
- Total Orders
- Average Order Value (AOV)
- Profit Margin

These KPIs provide a quick overview of overall business performance.

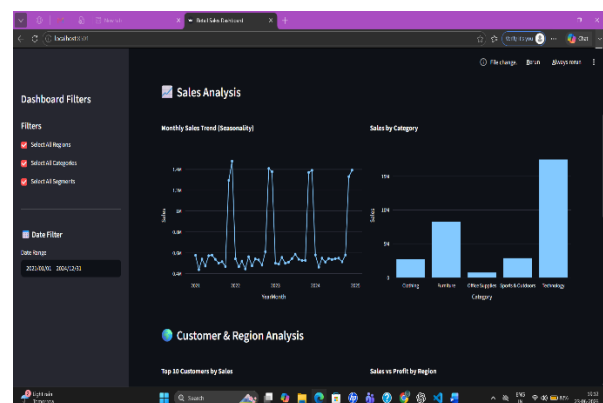
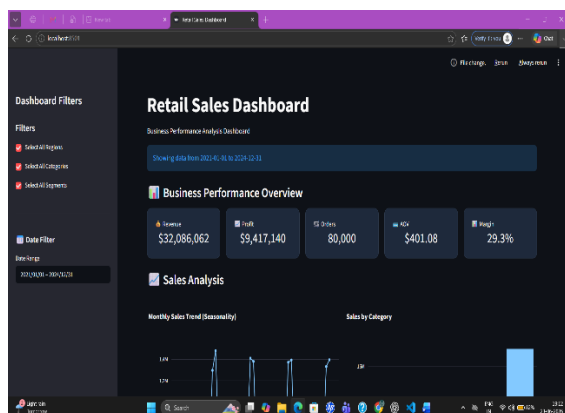
4. Interactive Dashboard Development

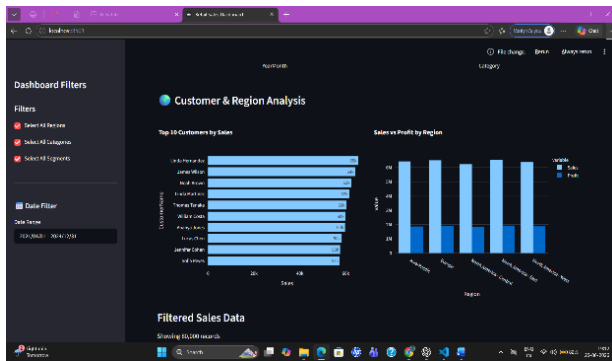
An interactive dashboard was developed using Streamlit and Plotly.

Dashboard Features:

- KPI Cards displaying Revenue, Profit, Orders, AOV, and Profit Margin.
- Region filter.
- Category filter.
- Segment filter.
- Date range filter.
- Monthly Sales Trend chart.
- Sales by Category chart.
- Top 10 Customers chart.
- Sales vs Profit by Region chart.
- Dynamic data table displaying filtered records.

All visualizations update automatically based on user-selected filters.





The dashboard displays a table of sales records. The table has columns for Order ID, Order Date, Ship Date, Ship Date, Channel, Customer ID, Customer Name, Order, Item, and Age. The data is filtered to show 10 records.

Order ID	Order Date	Ship Date	Ship Date	Channel	Customer ID	Customer Name	Order	Item	Age
010-1000001	2023-01-01 09:00:00	2023-01-01	2023-01-01	Standard	010-1000001	John Doe Inc	12.5M	John Doe Inc	1
010-1000002	2023-01-02 10:00:00	2023-01-02	2023-01-02	Standard	010-1000002	Jane Smith Corp	11.8M	Jane Smith Corp	2
010-1000003	2023-01-03 11:00:00	2023-01-03	2023-01-03	Standard	010-1000003	Global Innovations	10.2M	Global Innovations	3
010-1000004	2023-01-04 12:00:00	2023-01-04	2023-01-04	Standard	010-1000004	Future Tech Ltd	9.7M	Future Tech Ltd	4
010-1000005	2023-01-05 13:00:00	2023-01-05	2023-01-05	Standard	010-1000005	Enterprise Solutions	8.9M	Enterprise Solutions	5
010-1000006	2023-01-06 14:00:00	2023-01-06	2023-01-06	Standard	010-1000006	Prime Industries	8.1M	Prime Industries	6
010-1000007	2023-01-07 15:00:00	2023-01-07	2023-01-07	Standard	010-1000007	Velocity Corp	7.6M	Velocity Corp	7
010-1000008	2023-01-08 16:00:00	2023-01-08	2023-01-08	Standard	010-1000008	Apex Group	7.3M	Apex Group	8
010-1000009	2023-01-09 17:00:00	2023-01-09	2023-01-09	Standard	010-1000009	Quantum Dynamics	6.8M	Quantum Dynamics	9
010-1000010	2023-01-10 18:00:00	2023-01-10	2023-01-10	Standard	010-1000010	Starline Corp	6.5M	Starline Corp	10

Top 5 Insights

1. Strong Business Performance

The company generated approximately \$32.1 million in revenue and \$9.4 million in profit during the analysis period, demonstrating strong overall performance.

2. Healthy Profitability

The overall profit margin is approximately 29.3%, indicating efficient cost management and sustainable operations.

3. Seasonal Demand Patterns

Monthly sales trends show recurring peaks across multiple years, suggesting predictable seasonal demand cycles that can be leveraged for planning and forecasting.

4. Customer Revenue Concentration

A relatively small group of top customers contributes a significant portion of total revenue, highlighting the importance of customer retention.

5. Regional Performance Differences

Regional analysis shows differences in both sales and profitability, indicating opportunities to optimize strategies across locations.

Recommendations

1: Strengthen Seasonal Planning

Use historical sales trends to forecast demand, improve inventory management, and launch targeted campaigns before peak periods.

2: Focus on Customer Retention

Develop loyalty programs and personalized marketing initiatives to retain high-value customers and increase repeat purchases.

3: Improve Regional Performance

Investigate underperforming regions and implement targeted pricing, marketing, and inventory strategies to improve profitability.

Conclusion

This project successfully transformed raw sales data into actionable business intelligence through data cleaning, analysis, KPI tracking, and dashboard development. The interactive dashboard enables stakeholders to monitor performance, identify trends, and make data-driven decisions efficiently.